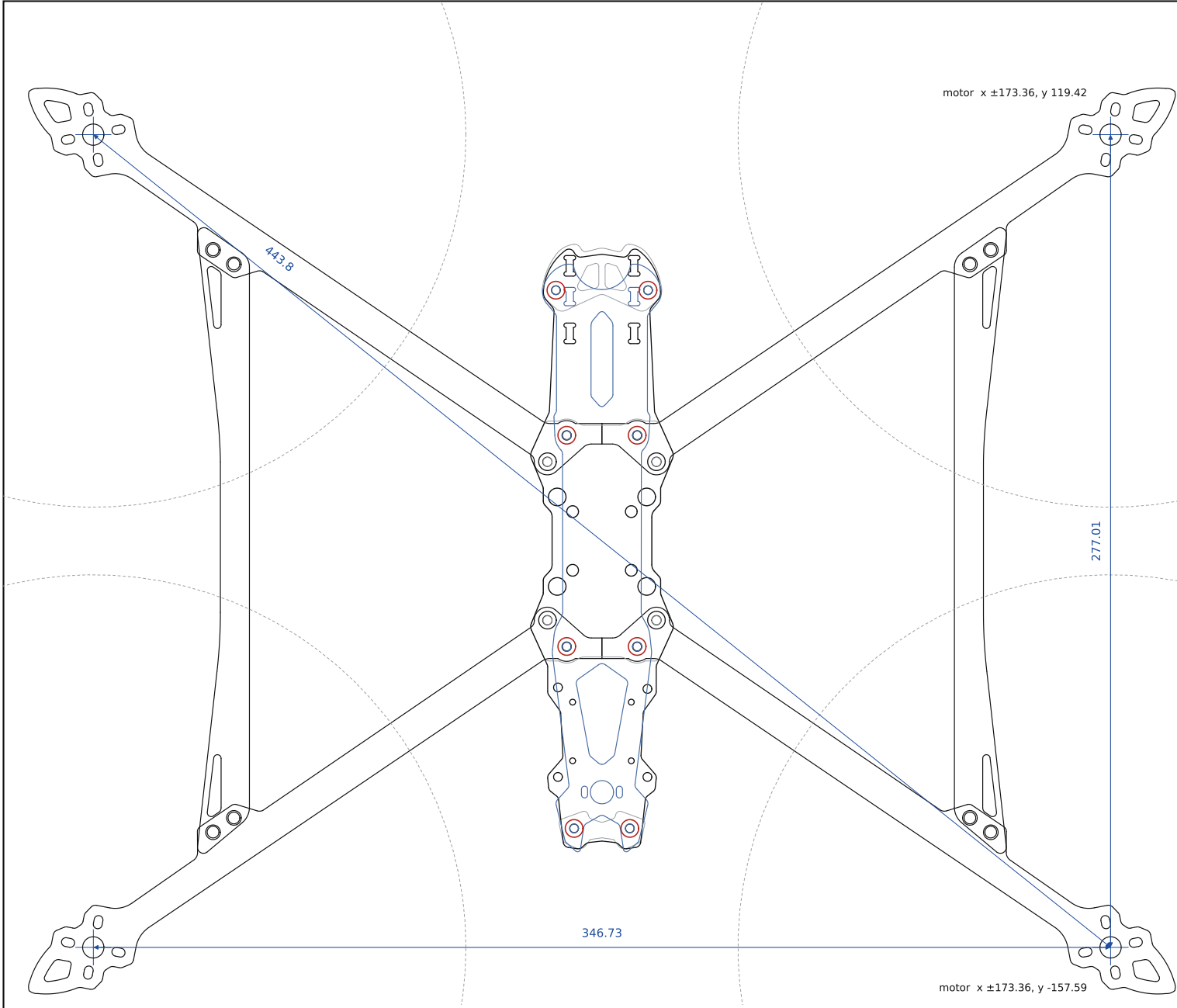




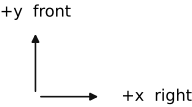
Parts

Qty	Part (STEP)	t mm	RJX manual
1	Top plate	2	Top plate
1	Lower plate	3	Mid plate
1	Belly plate	3	Bottom plate
1	Chin plate	2.25	Front plate
1	Tail plate	2.25	Back plate
4	Arm	7	Arm
2	Arm brace	2	Arm bracer
2	Camera plate	2.5	Camera plate
8	Standoff M3x35 Ø6		Standoff spacer

Stack-up: z in mm, 0 = underside of the lower plate

Part	z from	z to
top plate	38	40
standoffs	3	38
camera plates	0	40
lower plate	0	3
arm braces	0	2
chin, tail plates	-2.25	0
arms	-7	0
belly plate	-10	-7

Every part is a flat plate: extrude its DXF profile by its thickness, at its z.



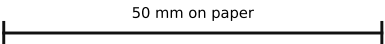
Seen from above, nose up.
Dashed: 10" prop discs.
Red: standoffs. Blue: top plate.

Frame overview

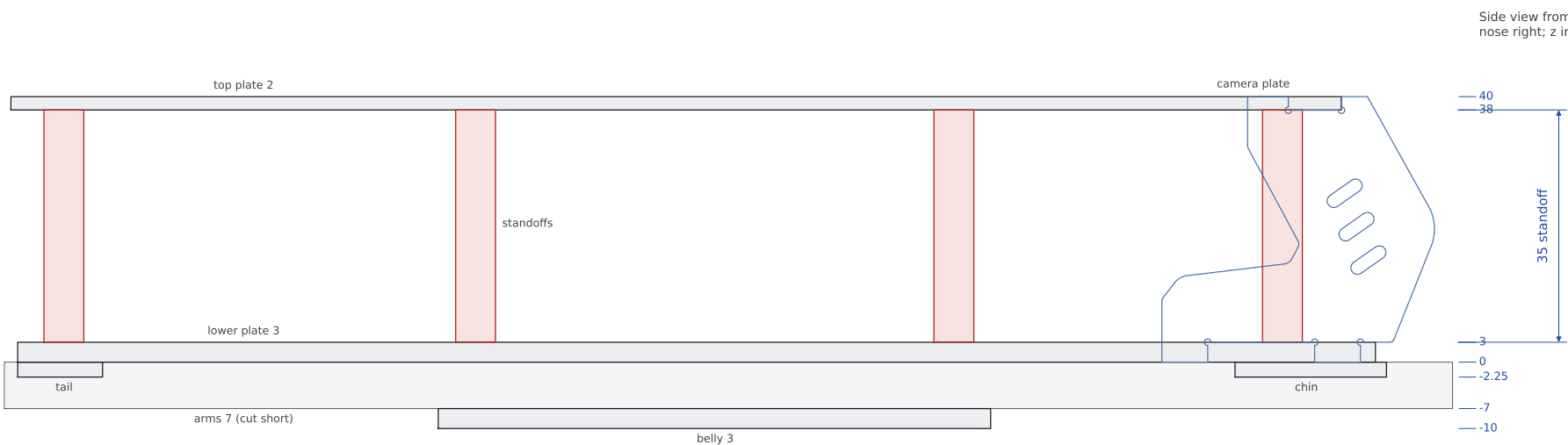
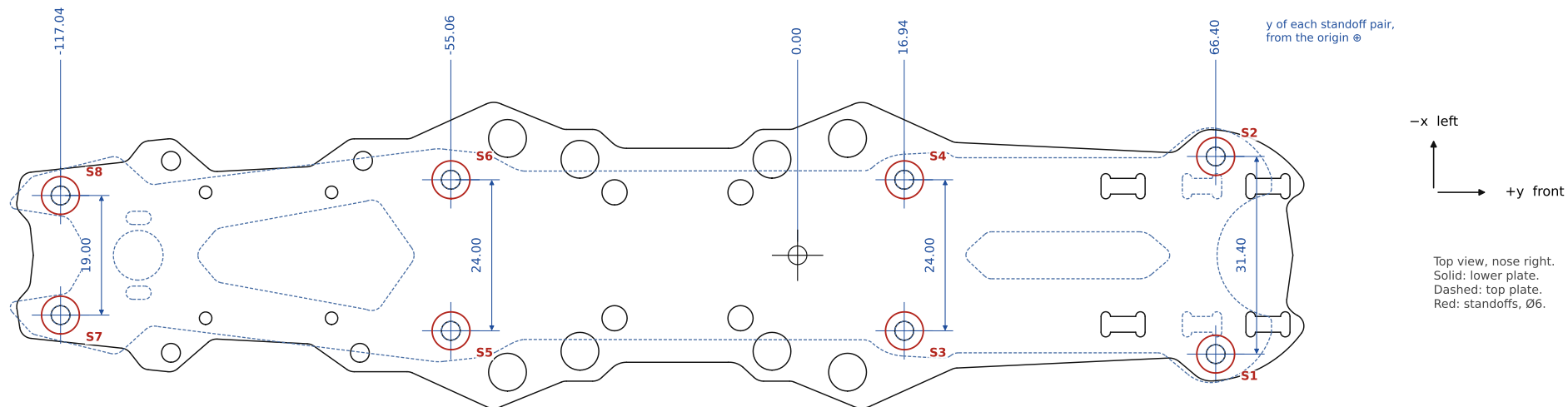
Scale 1:2

Motor centres 346.7 x 277.0 mm, 443.8 mm diagonal

Sheet 1 of 7



Mark4 V2 10" frame, from its STEP model. mm. Print at 100 % (actual size) and check the bar.

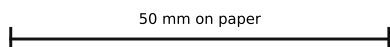


Standoff holes (lower plate; use these in every plate)

	x	y	Top plate	Bolts from below via
S1	15.70	66.40	same	Chin
S2	-15.70	66.40	same	Chin
S3	12.00	16.94	y 16.97	Arm, Belly
S4	-12.00	16.94	y 16.97	Arm, Belly

	x	y	Top plate	Bolts from below via
S5	12.00	-55.06	same	Arm, Belly
S6	-12.00	-55.06	same	Arm, Belly
S7	9.50	-117.04	same	Tail
S8	-9.50	-117.04	same	Tail

In the model every hole on a standoff axis is within 0.07 mm of it, and the top and lower plates within 0.03 mm of each other (CHECK.md).



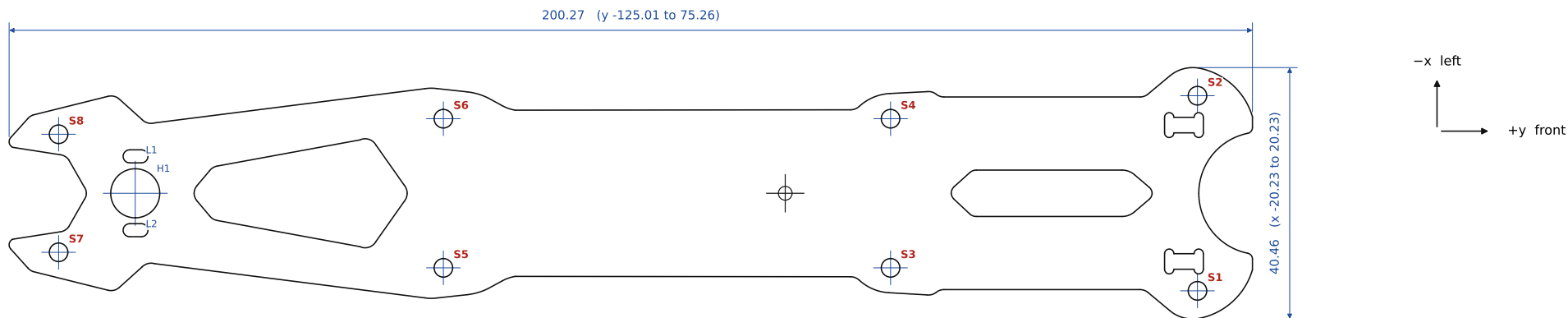
Mark4 V2 10" frame, from its STEP model. mm. Print at 100 % (actual size) and check the bar.

Standoffs (struts) and stack-up

8 x M3 x 35 mm standoffs, Ø6, lower plate to top plate

Scale 1:1

Sheet 2 of 7

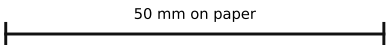


Hole centres from the model's origin @: x right, y forward. S1-S8: standoff holes, on the same axes in every plate. Thickness 2 mm.

4 cut-outs and the outline: take them from dxf/top_plate.dxf.

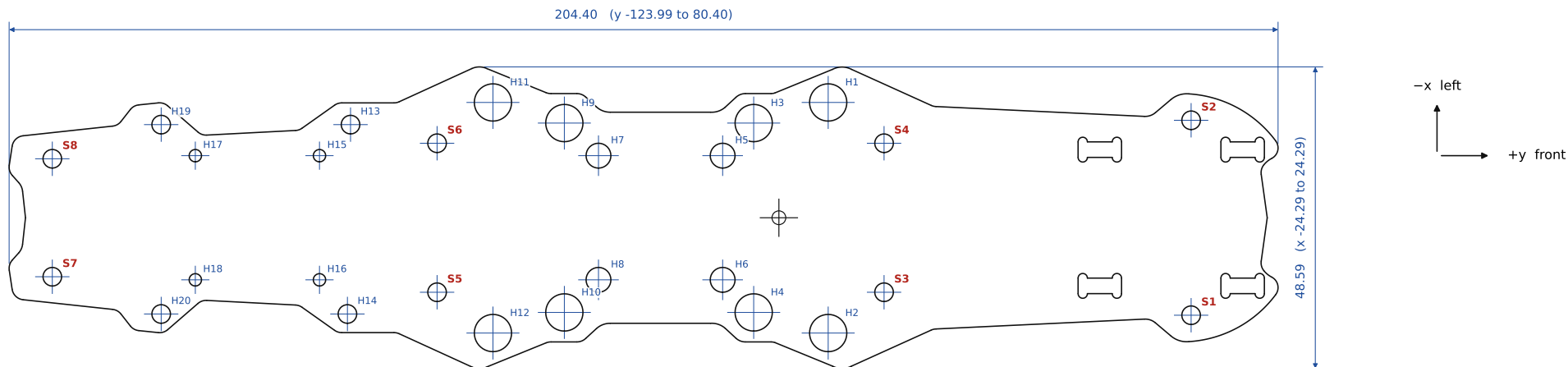
	x	y	Ø
S2	-15.70	66.40	3.00
S1	15.70	66.40	3.00
S4	-12.00	16.97	3.00
S3	12.00	16.97	3.00
S6	-12.00	-55.06	3.00
S5	12.00	-55.06	3.00
H1	0.00	-104.70	7.90
S8	-9.50	-117.04	3.00
S7	9.50	-117.04	3.00

Slot	x1	y1	x2	y2	w
L1	-5.94	-103.70	-5.94	-105.60	2.10
L2	5.94	-103.70	5.94	-105.60	2.10



Mark4 V2 10" frame, from its STEP model. mm. Print at 100 % (actual size) and check the bar.

Top plate	Scale 1:1
Qty 1. Carbon 2 mm. Battery on top	Sheet 3 of 7



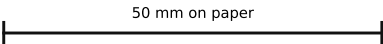
Hole centres from the model's origin @: x right, y forward. S1-S8: standoff holes, on the same axes in every plate. Thickness 3 mm.

4 cut-outs and the outline: take them from dxf/lower_plate.dxf.

	x	y	Ø
S2	-15.70	66.40	3.00
S1	15.70	66.40	3.00
S4	-12.00	16.94	3.00
S3	12.00	16.94	3.00
H1	-18.57	7.94	6.00
H2	18.57	7.94	6.00
H3	-15.25	-4.06	6.00
H4	15.25	-4.06	6.00
H5	-10.00	-9.06	4.00
H6	10.00	-9.06	4.00

	x	y	Ø
H7	-10.00	-29.06	4.00
H8	10.00	-29.06	4.00
H9	-15.25	-34.56	6.00
H10	15.25	-34.56	6.00
H11	-18.57	-46.06	6.00
H12	18.57	-46.06	6.00
S6	-12.00	-55.06	3.00
S5	12.00	-55.06	3.00
H13	-15.00	-69.00	3.00
H14	15.50	-69.50	3.00

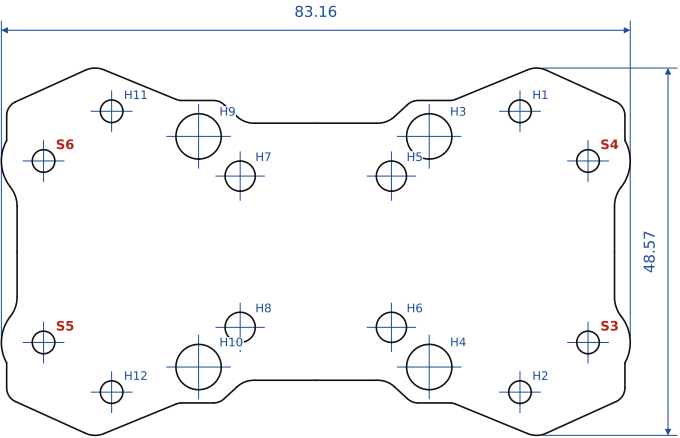
	x	y	Ø
H15	-10.00	-74.00	2.00
H16	10.00	-74.00	2.00
H17	-10.00	-94.00	2.00
H18	10.00	-94.00	2.00
H19	-15.00	-99.50	3.00
H20	15.50	-99.50	3.00
S8	-9.50	-117.04	3.00
S7	9.50	-117.04	3.00



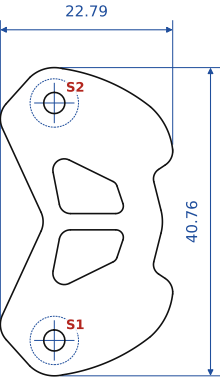
Mark4 V2 10" frame, from its STEP model. mm. Print at 100 % (actual size) and check the bar.

Lower plate (manual: mid plate)	Scale 1:1
Qty 1. Carbon 3 mm. Standoffs and the stacks	Sheet 4 of 7

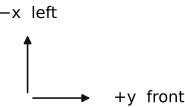
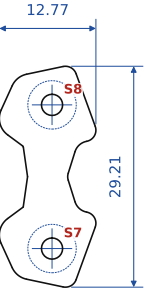
Belly plate



Chin plate



Tail plate



Belly plate

	x	y	Ø
S4	-12.00	16.90	3.00
S3	12.00	16.90	3.00
H1	-18.57	7.90	3.00
H2	18.57	7.90	3.00
H3	-15.25	-4.10	6.00
H4	15.25	-4.10	6.00
H5	-10.00	-9.10	4.00
H6	10.00	-9.10	4.00

	x	y	Ø
H7	-10.00	-29.10	4.00
H8	10.00	-29.10	4.00
H9	-15.25	-34.60	6.00
H10	15.25	-34.60	6.00
H11	-18.57	-46.10	3.00
H12	18.57	-46.10	3.00
S6	-12.00	-55.10	3.00
S5	12.00	-55.10	3.00

Chin plate

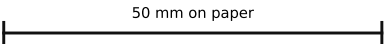
	x	y	Ø	
S2	-15.70	66.40	2.80	csk Ø6.4 x 90°, under
S1	15.70	66.40	2.80	csk Ø6.4 x 90°, under

Tail plate

	x	y	Ø	
S8	-9.50	-117.01	2.80	csk Ø6.4 x 90°, under
S7	9.50	-117.01	2.80	csk Ø6.4 x 90°, under

Chin and tail: Ø2.8 through, countersunk (dashed) on the underside, under standoffs S1, S2 and S7, S8.

Belly: under the arms; S3-S6 bolt up through it, the arms and the lower plate into the standoffs. Hole centres from the model's origin.



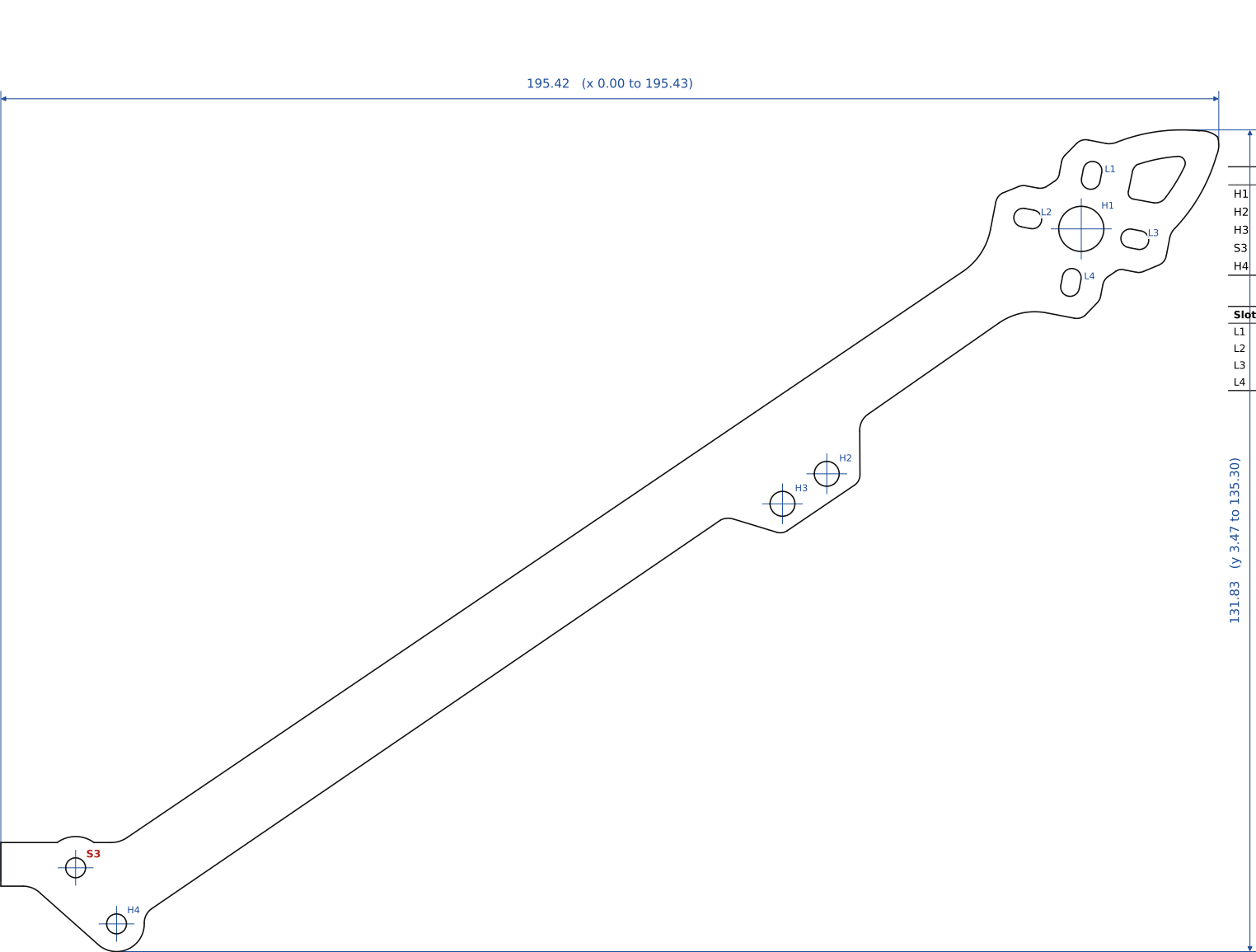
Mark4 V2 10" frame, from its STEP model. mm. Print at 100 % (actual size) and check the bar.

Belly, chin and tail plates

Scale 1:1

1 each. Belly carbon 3 mm; chin and tail 2.25 mm

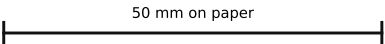
Sheet 5 of 7



	x	y	Ø
H1	173.36	119.42	7.25
H2	132.53	80.13	4.00
H3	125.42	75.31	4.00
S3	12.01	16.92	3.20
H4	18.58	7.92	3.20

Slot	x1	y1	x2	y2	w
L1	174.88	127.28	175.17	128.75	3.00
L2	164.04	121.23	165.51	120.94	3.00
L3	181.22	117.90	182.69	117.62	3.00
L4	171.56	110.09	171.84	111.57	3.00

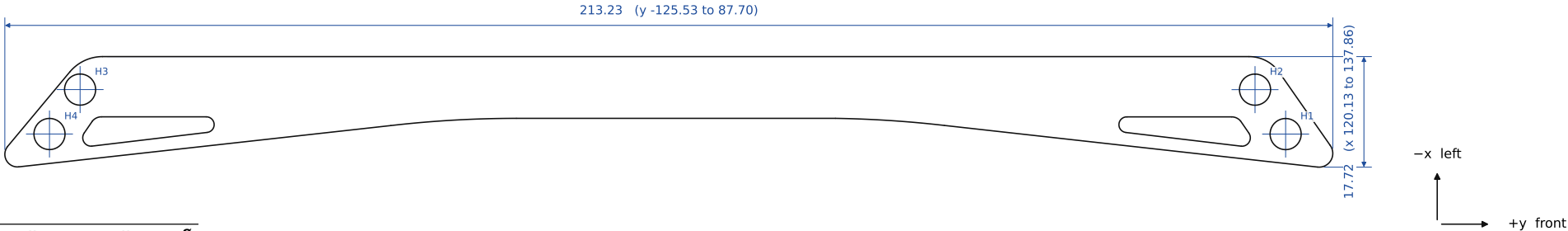
Motor centre H1: x 173.36, y 119.42, Ø7.25. The motor slots run from radius 8.00 to 9.50 about it: bolt circles Ø16.0 to Ø19.0 (16x16 and 19x19 motors).
S3 (and S4-S6 on the mirrored arms) carries a standoff bolt. The outline and the 1 lightening cut-out come from dxf/arm.dxf.



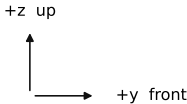
Mark4 V2 10" frame, from its STEP model. mm. Print at 100 % (actual size) and check the bar.

Arm	Scale 1:1
Qty 4. Carbon 7 mm. Front right drawn; the others are mirror images	Sheet 6 of 7

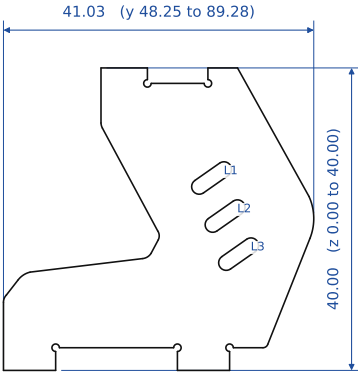
Arm brace (right), from above



	x	y	Ø
H1	132.55	80.13	5.00
H2	125.43	75.22	5.00
H3	125.42	-113.45	5.00
H4	132.54	-118.36	5.00



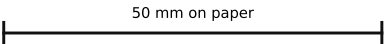
Camera plate (right), from the right



Slot	y1	z1	y2	z2	w
L1	74.13	24.30	77.41	26.59	2.00
L2	75.92	19.26	79.19	21.56	2.00
L3	77.70	14.23	80.98	16.52	2.00

Camera plates: z 0 to 40. Their tabs pass through the slots in the lower and top plates, flush with the outside faces. The right one's faces are at x 9.72 and 12.23.

Brace: on top of the arms, z 0 to 2; its holes line up with the arm's H2 and H3.



Mark4 V2 10" frame, from its STEP model. mm. Print at 100 % (actual size) and check the bar.

Arm brace, camera plate	Scale 1:1
Brace qty 2, 2 mm. Camera plate qty 2, 2.5 mm	Sheet 7 of 7